

October 8, 2026

The Editors
Knowledge and Information Systems
Springer

Re: Submission of an original research article — “Sieve: Budgeted Question-Conditioned Information Selection for Training-Free Long-Context Compression”

Dear Editors,

Please consider the manuscript referenced above for publication in *Knowledge and Information Systems* as an original research article. I am the sole author of the manuscript and serve as its corresponding author.

I believe the manuscript fits the scope of *Knowledge and Information Systems* closely. The study is a quantitative treatment of question-conditioned information selection for long-context question answering, built directly on classical information-retrieval machinery — IDF weighting, BM25 ranking, and maximal marginal relevance — and on the data-engineering question of how a knowledge system should budget its context: which sentences to keep under a fixed word budget, at what selection quality, and at what computational cost. Every method is evaluated with recall protocols over annotated evidence, and quality, compression, and compressor cost are reported together so that practitioners of information systems have a measured basis for deciding when statistical selection suffices and when a hosted scorer earns its infrastructure. This is, in the journal’s terms, an enabling-technology study for knowledge and information processing: a training-free, model-free component that any long-context pipeline can compose with its existing retrieval stack.

The manuscript formulates long-context compression as *budgeted question-conditioned information selection*: given a fixed word budget, choose the subset of sentences that jointly serves question-relevance coverage, positional reliability, and diversity, with the selector’s own computational cost treated as a first-class constraint rather than an afterthought. The formulation is stated formally, and a thirteen-entry design-space table locates every prior selector along seven dimensions (question conditioning, positional awareness, redundancy control, model-free inference, training-free operation, sentence-level intact units, and explicit budget with measured cost). The corner that table leaves unoccupied — to our knowledge, and stated with the audit’s scope conditions — is the one this paper measures: a selector that is simultaneously question-conditioned, position-aware, redundancy-controlling, and entirely model-free.

The vehicle of the study is *Sieve*, the instantiation of that formulation in this corner: IDF-weighted coverage of query content terms, a U-shaped positional prior, and a redundancy penalty applied greedily under the explicit budget. Its preprocessing is a linear-time pass over the context, its budgeted selection is $O(k n \bar{s})$, and the full selector runs in tens of milliseconds on one CPU core (50.9–54.3 ms at a 6,000-word context), with no model parameters, no training, and no dependency beyond NumPy.

The evaluation is organized as a three-way quality–compression–cost frontier study over three evidence-annotated benchmarks — QASPER, HotpotQA (distractor setting), and 2WikiMultihopQA, 1,600 records in total. *Sieve* retains more gold evidence than every question-agnostic baseline on all three datasets (margins over head truncation of 12.8, 44.8, and 22.8 recall points at 4 $\times$), while BM25 sentence ranking, the strongest question-conditioned model-free comparator, retains 43.1% against *Sieve*’s 36.2% on QASPER — a result the manuscript reports as measured and reads through its own ablations, where the relevance term alone recovers 43.6% against 36.2% for the full configuration, as a structural trade of raw recall for positional reliability. A dedicated redundancy analysis shows the penalty is a measured property rather than a design assumption: removing it raises selected-set self-similarity and duplicate content-token rate on every dataset–budget combination (paired intervals excluding zero). The end-task protocol runs seven selection methods at $n=100$ each under a production LLM reader: the BM25 reference retains 76.0% of full-context F1 against 61.8% for *Sieve* (paired gap 4.8 points, interval excluding zero), while *Sieve* remains statistically matched with truncation (paired gap +4.0, interval crossing zero) — the same trade-off, consistent at both the selection and the answer layer. A GPT-2 perplexity gate, measured on the same hardware as every other method, reaches 28.7% evidence recall at 255–320 $\times$ *Sieve*’s measured cost; and the correlation between evidence recall and end-task F1 is weak (pooled Pearson 0.22 over 600 pairs), so the two protocols are reported separately throughout.

The manuscript substantially extends an earlier conference-format study of the same method; the statement of extensions appears in the Introduction. The extensions include the three-dataset evaluation, the BM25 baseline, the same-hardware comparison against a neural perplexity gate, the formal layer (formulation, propositions, cost statements), the recall–F1 correlation study, the end-to-end seven-method protocol at $n=100$, the configuration–budget–dataset ablation matrix, the reader-side positional profile, and the measured redundancy analysis.

I confirm that: the manuscript is original and has not been published previously in any form; it is not under consideration by any other journal; I have no known competing financial interests or personal relationships that could have appeared to influence the work reported; and the work received no specific funding. The benchmarks used are public (QASPER, HotpotQA, 2WikiMultihopQA), and the released repository reproduces every table and figure from raw checkpoints.

Thank you for considering the manuscript. I look forward to the reviewers’ comments.

Sincerely,

Chang Tan

Undergraduate Student, School of Economics and Management

Dalian University of Technology

No. 2 Linggong Road, Ganjingzi District, Dalian, Liaoning 116024, China

E-mail: aquars43@foxmail.com